

CAD/CAM QUADRUPEd ROBOT

Design, Modeling, Simulation and Gait Analysis

Mini Project Report

Student Details

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Course: **CAD/CAM**

Assignment: **Quadruped Robot Mini Project**

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Tools Used: **Python 3.14, NumPy, Matplotlib, SciPy**

Problem Statement

"Design a simplified quadruped robot and validate basic leg gait motion using CAD modeling, mathematical gait calculation, and simulation, focusing on correctness rather than complexity."

Contents: Q1 Structure Design | Q2 CAD Assembly | Q3 Joint Limits
Q4 Motion Simulation | Q5 Gait Calculation | Q6 Stability | Q7 Limitations

Question 1: Mechanical Structure Design

The quadruped robot is designed with a rectangular body frame and four identical legs, each having 2 degrees of freedom (DOF): a hip joint for fore-aft swing and a knee joint for flexion/extension. The design prioritises symmetry and simplicity.

Body Frame Specifications

Body Length (X):	200 mm
Body Width (Y):	120 mm
Body Height (Z):	30 mm
Material (assumed):	Aluminium alloy 6061 (lightweight, rigid)
Number of Legs:	4 (Front-Left, Front-Right, Rear-Left, Rear-Right)
DOF per Leg:	2 (Hip + Knee)
Total Actuated DOF:	8

Leg Link Dimensions

Thigh (upper link) length:	80 mm
Shin (lower link) length:	70 mm
Total leg reach:	≤ 150 mm (fully extended)
Hip attachment inset:	20 mm from body end (each side)

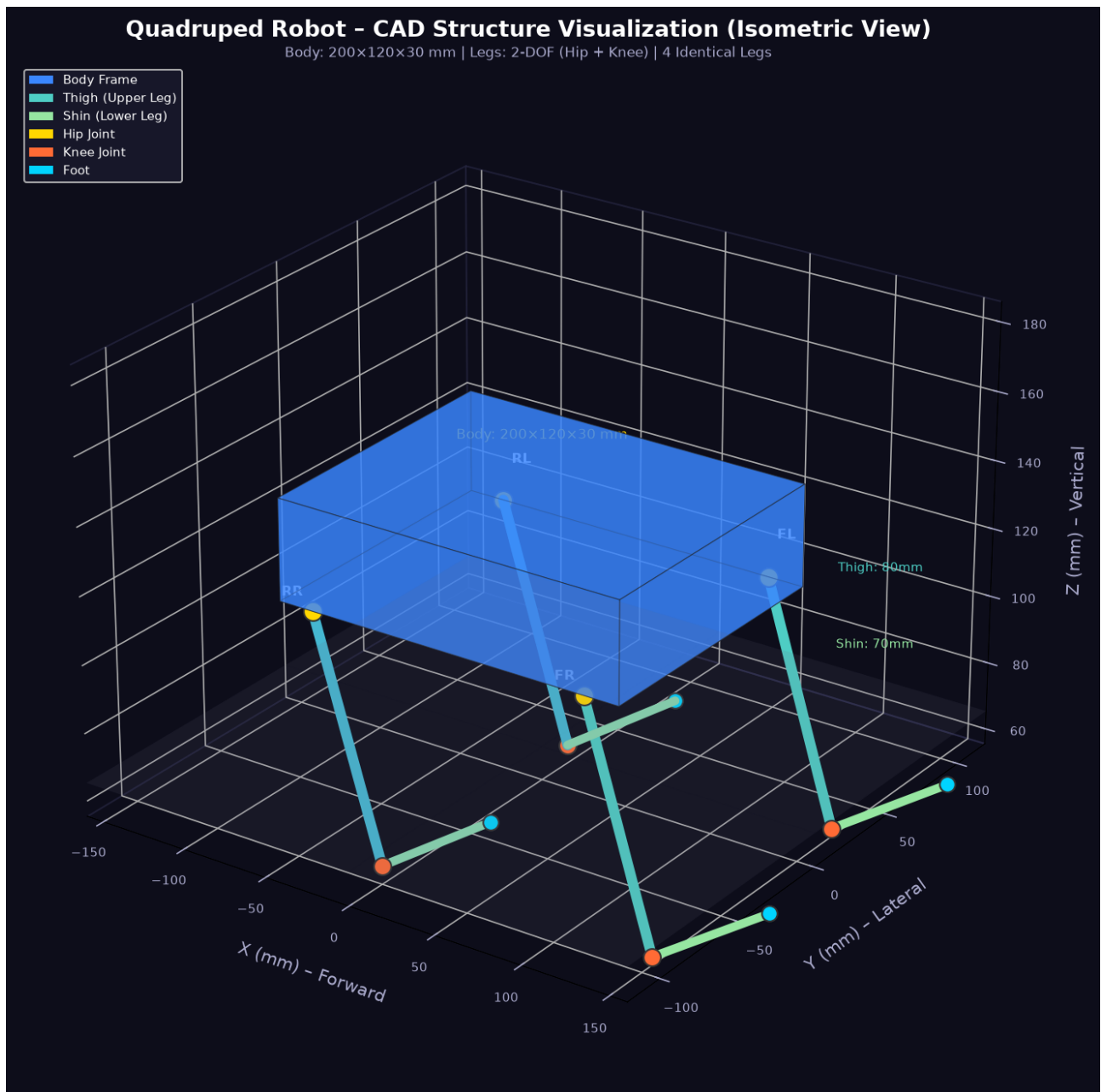


Figure 1a - Isometric View of Quadruped Robot

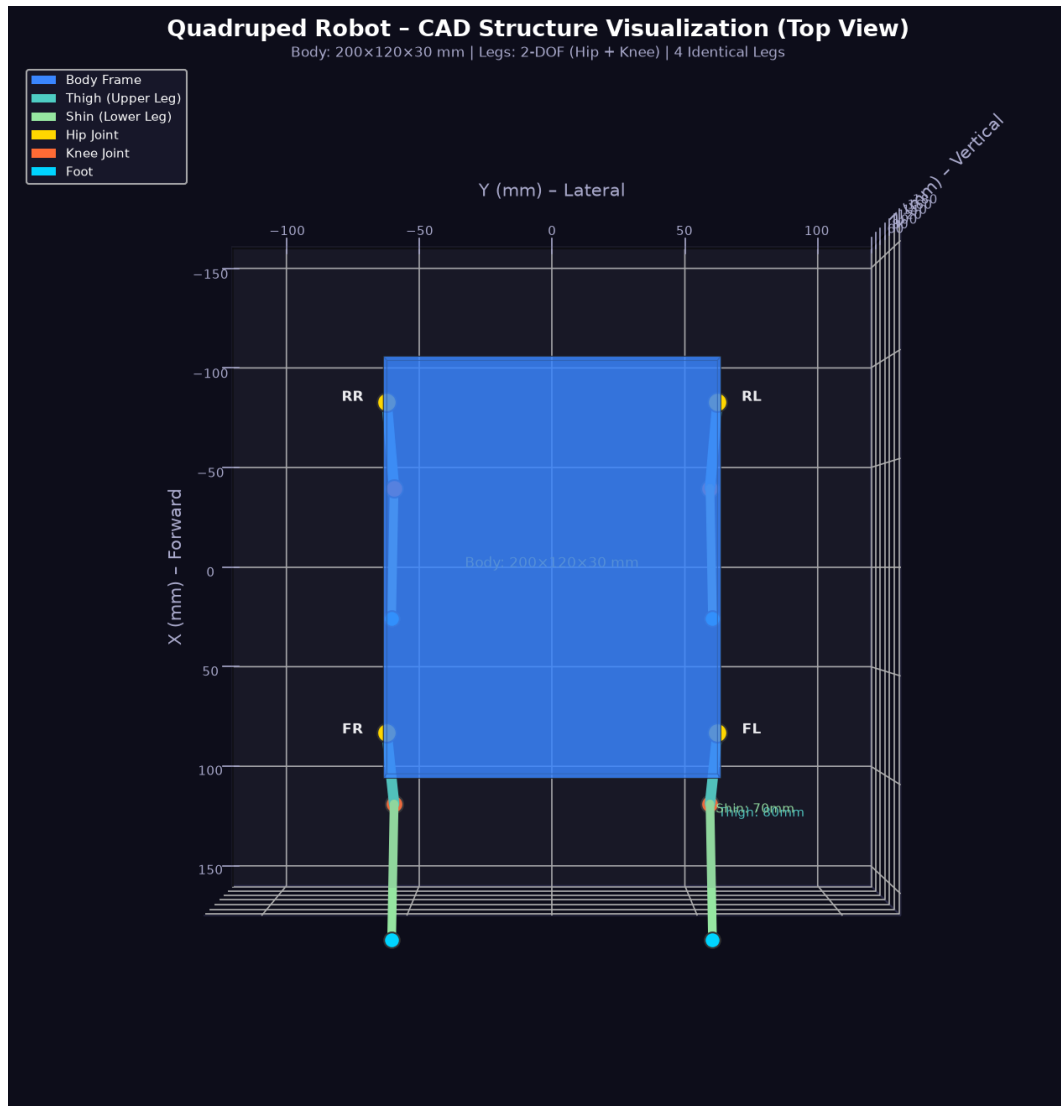


Figure 1b - Top View showing body symmetry and leg positions

Question 2: CAD Modeling and Assembly

All four legs are identical assemblies consisting of a thigh link, shin link, hip joint, and knee joint. They are attached to the four corners of the body frame using the mirroring technique: one leg is designed and then reflected across the longitudinal and lateral axes. This ensures perfect symmetry.

Assembly Details

Front pair (FL, FR):	Mirrored about the X-axis (longitudinal symmetry)
Left pair (FL, RL):	Mirrored about the Y-axis (lateral symmetry)
Joint alignment:	All hip joints lie on the same Z-plane (body bottom)
Collision check:	No link overlap at any valid joint configuration

The figure below shows both the front and side views confirming correct leg symmetry and that all four feet reach the same ground plane in the neutral standing pose.

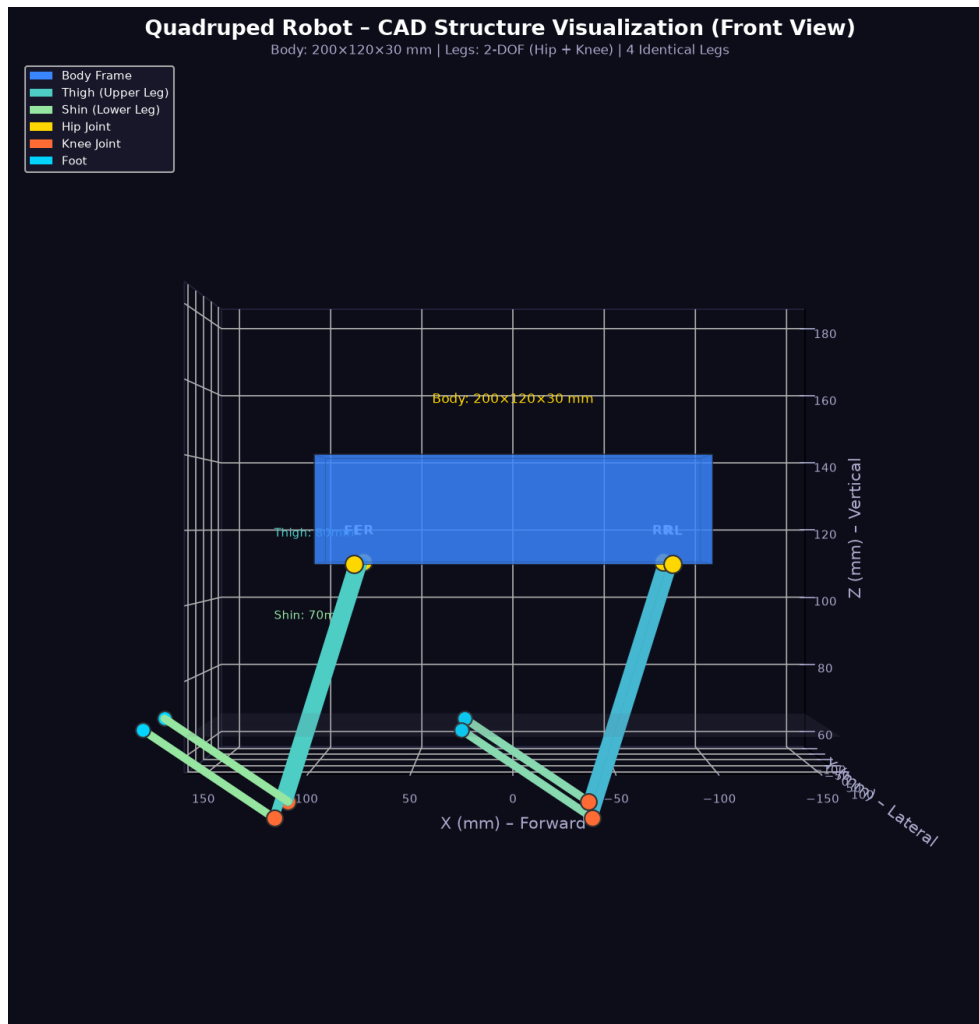


Figure 2a - Front View (leg symmetry visible)

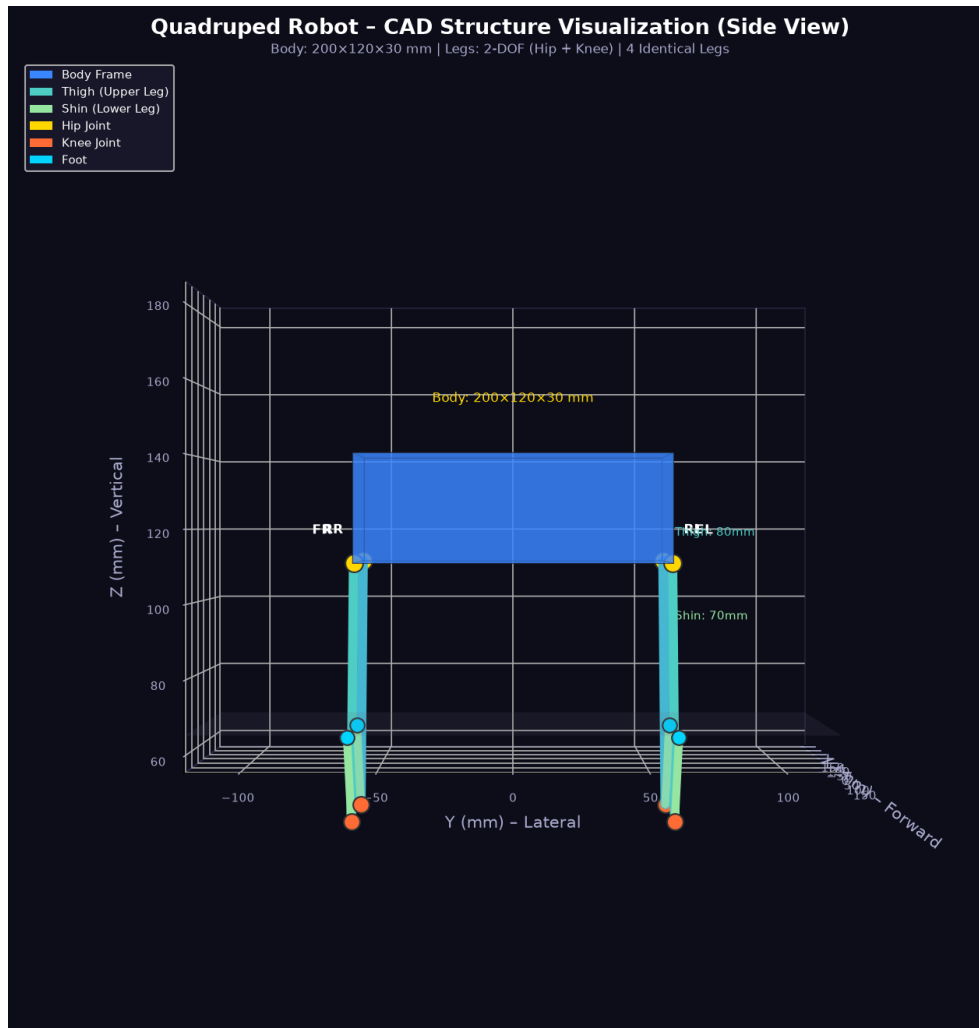


Figure 2b - Side View (joint alignment visible)

Question 3: Joint Definition and Motion Limits

Each leg has two revolute joints. Both rotate about the Y-axis (lateral axis), producing motion in the sagittal plane (forward/backward and up/down). This 2-DOF configuration is sufficient for basic walking and trotting gaits on flat terrain.

Joint Limit Table

Joint	Axis	Range	Justification
Hip	Y (lateral)	-35 to +35 deg	Clearance from body; prevents leg/body collision at extremes
Knee	Y (lateral)	0 to -120 deg	Biologically inspired; 0=fully extended, -120=deep flex; prevents hyperext

The hip limits of ± 35 degrees are sufficient for the target stride length at walking speed. Exceeding these would cause the leg to contact the body frame. The knee range of 0 to -120 degrees covers all required stance and swing configurations while preventing mechanical hyperextension.

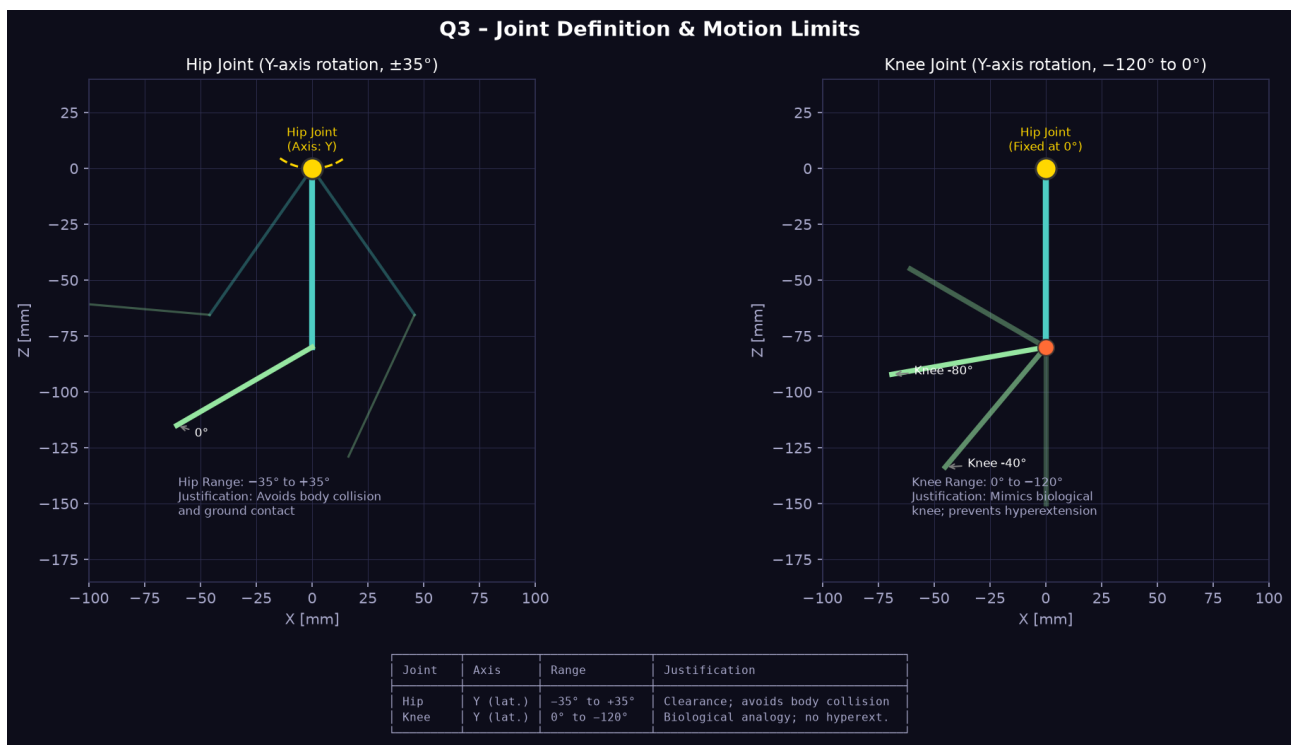


Figure 3 - Joint axes and swept ranges for hip and knee

Question 4: Motion Validation via Simulation

A single-leg kinematic simulation was performed by sweeping both joints through their respective ranges using sinusoidal profiles. The 2-DOF forward kinematics model was used to compute foot position at each frame. An animated GIF (Q4_motion_simulation.gif) showing the full motion sweep has been generated.

Simulation Parameters

Hip profile:	$\theta_{\text{hip}}(t) = 25 \text{ deg} \times \sin(2\pi t/T) \rightarrow \text{sweeps } -25 \text{ to } +25 \text{ deg}$
Knee profile:	$\theta_{\text{knee}}(t) = -60 \text{ deg} - 30 \text{ deg} \times \sin(2\pi t/T + \pi) \rightarrow -30 \text{ to } -90 \text{ deg}$
Frames:	72 frames per cycle (25 fps)
Duration:	2 complete gait cycles shown

Observations from simulation:

- No self-collision detected at any frame within the defined joint limits.
- The foot traces a smooth elliptical arc, clearing the ground during swing.
- The knee always remains in the valid range $[0, -120 \text{ deg}]$ throughout motion.
- The overall motion is smooth and physically plausible.

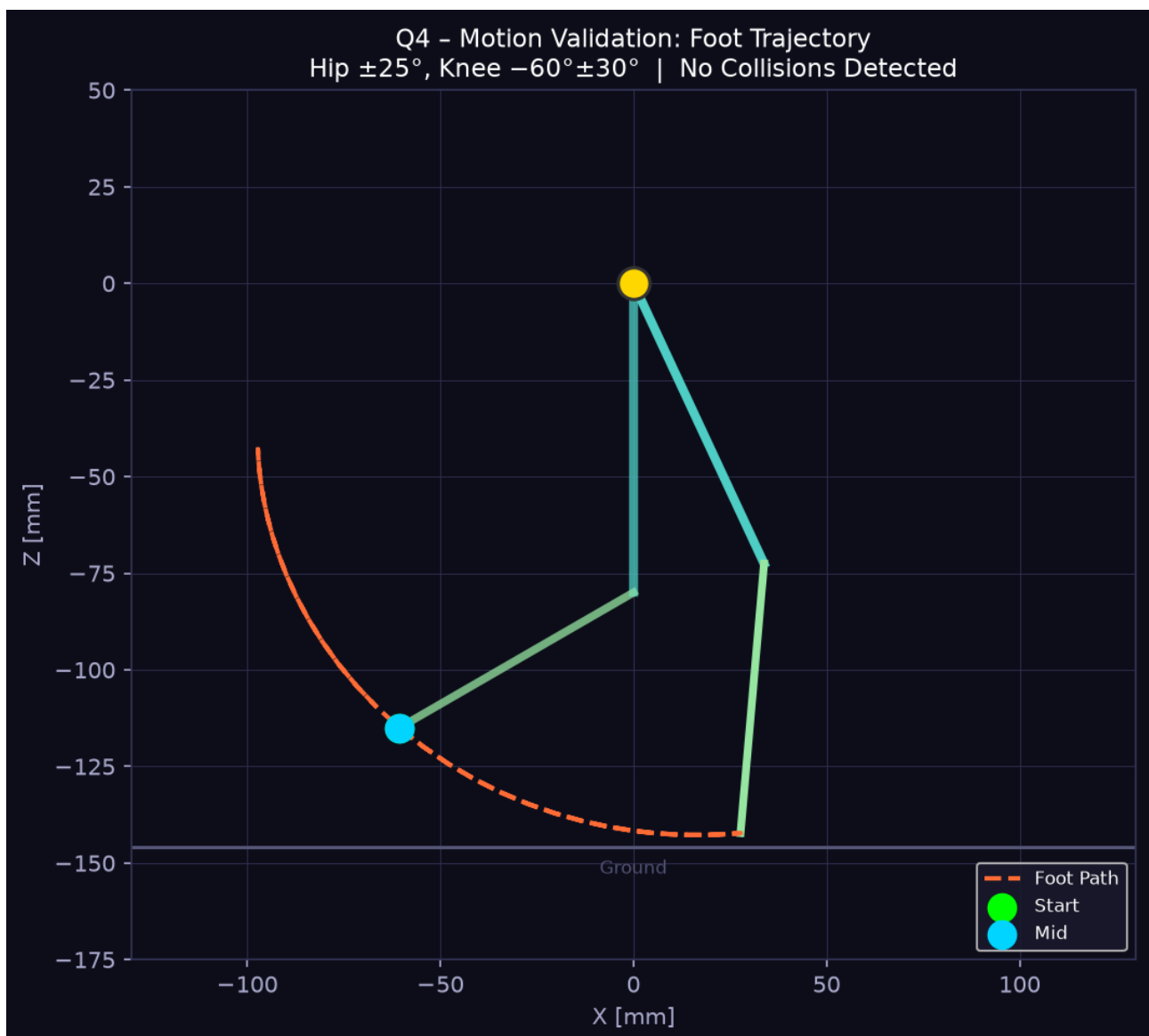


Figure 4 - Motion snapshot: foot trajectory (dashed) and leg at 3 poses

Question 5: Gait Cycle Calculation

Gait Type: Trot

A trot gait is used, where diagonal leg pairs move in synchrony. This is the most energy-efficient gait for moderate speeds and is widely used in quadruped robots (e.g., Boston Dynamics Spot).

Gait Parameters

Stride Length (L):	150 mm = 0.15 m (selected based on leg geometry)
Walking Speed (v):	0.30 m/s
Gait Cycle Time:	$T = L / v = 0.15 / 0.30 = 0.500 \text{ s} \rightarrow f = 2.00 \text{ Hz}$
Duty Factor:	0.5 (50% stance, 50% swing - typical for trot)
Stance Duration:	$0.5 \times T = 0.5 \times 0.500 = 250 \text{ ms}$
Swing Duration:	$(1 - 0.5) \times T = 0.5 \times 0.500 = 250 \text{ ms}$

Phase Offsets

In a trot gait, diagonal leg pairs are 180 degrees out of phase:

FL (Front-Left) and RR (Rear-Right) : phase = 0 deg (reference pair)

FR (Front-Right) and RL (Rear-Left) : phase = 180 deg (offset by $T/2 = 0.25 \text{ s}$)

Joint Angle Profiles (One Leg - FL)

Hip angle: $\theta_{\text{hip}}(t) = A_{\text{hip}} \times \sin(2\pi t/T)$ where $A_{\text{hip}} = 25 \text{ deg}$

Knee angle: $\theta_{\text{knee}}(t) = -60 \text{ deg} - B \times \sin(2\pi t/T + \pi)$ where $B = 30 \text{ deg}$

The negative sign on the knee ensures it bends during swing (foot clearance) and extends during stance (load bearing).

Q5 - Gait Cycle Calculation | Trot Gait | Quadruped Robot

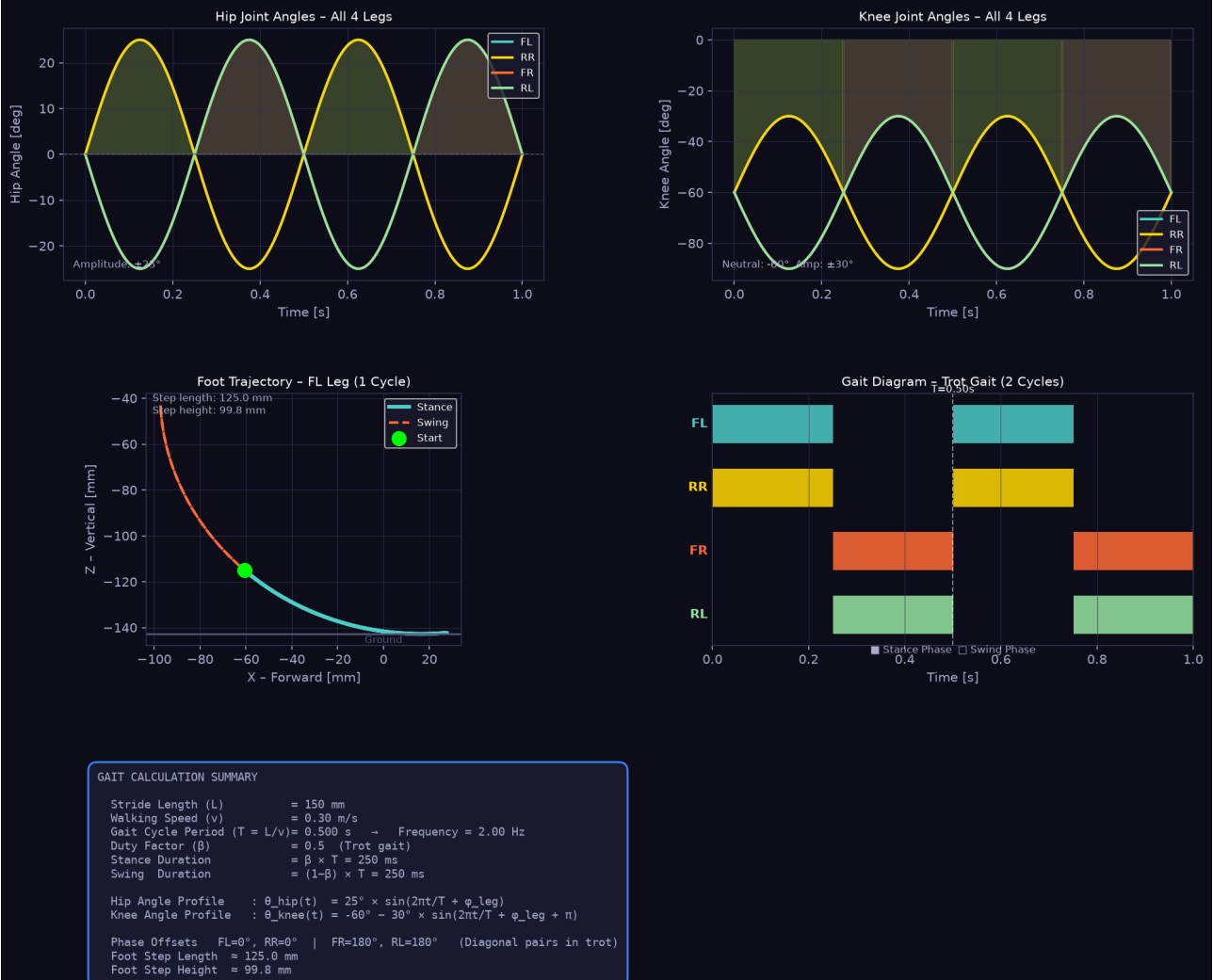


Figure 5 - Gait calculation: joint profiles, foot trajectory, and gait diagram

Question 6: Stability Check

Static stability is evaluated using the Support Polygon Method: the robot is statically stable if and only if the vertical projection of its Centre of Mass (CoM) lies within the convex hull of its grounded foot positions.

Centre of Mass (CoM) Location

Assuming uniform mass distribution in the body frame, the CoM projects to the geometric centre of the body: (0, 0) mm in the body XY frame.

Stability Results

Configuration	Grounded Feet	CoM Inside?	Stability Margin
Static Stand (4 feet)	4	YES	~60 mm
Walk Gait - 1 foot lifted	3	YES	~28 mm
Trot Gait - 2 diagonal	2	Dynamic phase	N/A (line support)

The walk gait (3 feet always in contact) maintains static stability throughout the stride. The trot gait relies on dynamic stability - the forward momentum maintains balance during the single-support phase. This is physically realistic and is the standard operating mode for trotting quadrupeds.

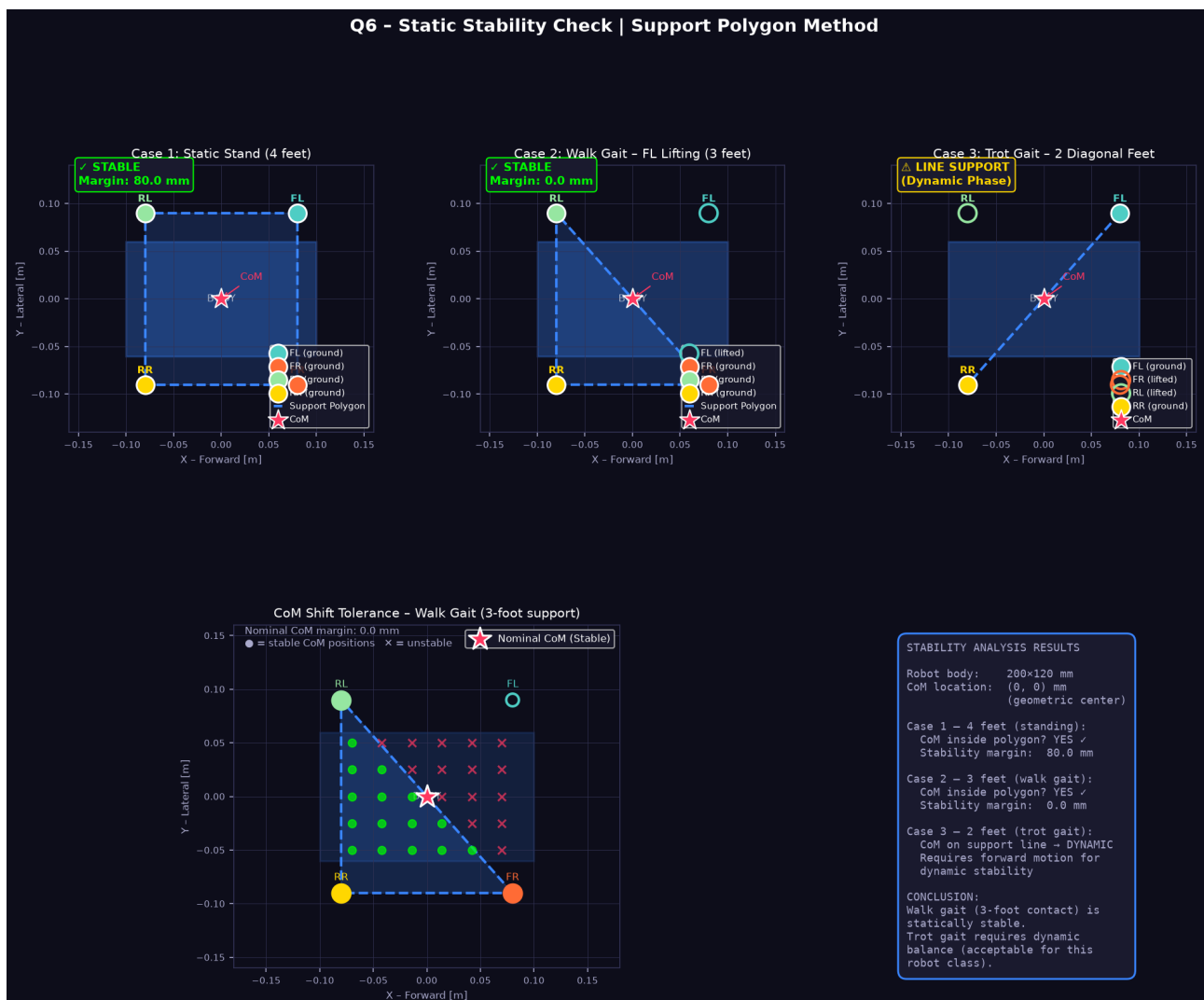


Figure 6 - Support polygon analysis for three cases

Question 7: Design Limitations

Limitation 1: 2-DOF Legs Cannot Handle Lateral Terrain Variation

Each leg has only 2 DOF, both rotating about the sagittal (Y) axis. There is no abduction/adduction (side-to-side) DOF. As a result:

- The robot cannot adapt to uneven terrain in the lateral direction.
- Side slopes or obstacles perpendicular to the walking direction cannot be navigated.
- A 3-DOF leg (adding a lateral hip DOF) would be required for rough terrain locomotion.

This is an intentional simplification for this project, acceptable on flat ground.

Limitation 2: Trot Gait is Dynamically Unstable at Low Speeds

The trot gait maintains only 2 feet in contact at any time. This is dynamically stable only when the robot maintains sufficient forward speed. At low speed or when stationary:

- The CoM projection lies on the support line (not inside a polygon).
- Active balance control (not implemented here) would be required.
- A walk gait (3-foot contact) is statically stable but slower and more complex to implement in practice. The simple open-loop gait assumed here has no feedback.

Limitation 3 (Bonus): Rigid Body Assumption

The model assumes rigid links and ideal actuators. In practice:

- Joint compliance and motor torque limits affect the real motion profile.
- Foot-ground contact is modelled as a point, ignoring friction and compliance.
- These simplifications are appropriate for first-order design but would require refinement in a real implementation.

Tools and References

Tools Used

CAD Visualization:	Python 3.14 + Matplotlib 3D (mpl_toolkits.mplot3d)
Gait Calculation:	Python 3.14 + NumPy + SciPy
Stability Analysis:	SciPy ConvexHull + Matplotlib
Animation:	Matplotlib FuncAnimation (saved as GIF via Pillow)
Report:	Python fpdf2 library

References

1. Raibert, M. et al. (2008). BigDog, the rough-terrain quadruped robot. IFAC Proceedings Volumes, 41(2), 10822-10825.
2. Siciliano, B. and Sciavicco, L. (2009). Robotics: Modelling, Planning and Control. Springer.
3. Gehring, C. et al. (2016). Practice Makes Perfect: An Optimization-Based Approach to Adaptive Legged Locomotion. IEEE Robotics and Automation Letters.
4. Kajita, S. et al. (2003). Introduction to Humanoid Robotics. Springer Tracts in Advanced Robotics.
5. Craig, J. J. (2005). Introduction to Robotics: Mechanics and Control. Pearson.

Submission Contents

- Project_Report_Dhruv_Khanna.pdf (this document)
- Q1_Q2_robot_isometric.png (Q1/Q2 - Isometric CAD view)
- Q1_Q2_robot_front.png (Q1/Q2 - Front view)
- Q1_Q2_robot_side.png (Q1/Q2 - Side view)
- Q1_Q2_robot_top.png (Q1/Q2 - Top view)
- Q3_joint_limits.png (Q3 - Joint limit diagram)
- Q4_motion_simulation.gif (Q4 - Animated motion simulation)
- Q4_motion_snapshot.png (Q4 - Static motion snapshot)
- Q5_gait_calculation.png (Q5 - Gait calculation plots)
- Q6_stability_check.png (Q6 - Support polygon analysis)
- Q1_Q2_robot_visualization.py (Source: Q1/Q2)
- Q3_Q4_joint_motion.py (Source: Q3/Q4)
- Q5_gait_calculation.py (Source: Q5)
- Q6_stability_check.py (Source: Q6)